
MA 6040 - Fuzzy Logic Connectives: Theory and Applications
Assignment 3 - Fuzzy Conjunctions - 27, September 2026

Consider the following t-norms for this assignment:

$$\begin{aligned} T_{\mathbf{nM}}(x, y) &= \begin{cases} 0 & \text{if } x + y \leq 1 \\ \min\{x, y\} & \text{if } x + y > 1 \end{cases} & T_{\mathbf{LK}}(x, y) &= \max\{0, x + y - 1\} \\ T_{\mathbf{M}}(x, y) &= \min\{x, y\} & T_{\mathbf{P}}(x, y) &= xy \\ T_{\mathbf{D}}(x, y) &= \begin{cases} \min\{x, y\}, & \text{if } \max\{x, y\} = 1 \\ 0, & \text{otherwise} \end{cases} & T_{\mathbf{MD}}(x, y) &= \begin{cases} 0, & \text{if } (x, y) \in (0, .5)^2 \\ \min\{x, y\}, & \text{otherwise} \end{cases} \end{aligned}$$

1. **Natural negation of a t-norm** is defined as follows

$$N_T(x) = \sup\{t \in [0, 1] \mid T(x, t) = 0\}.$$

Find the natural negation of the above t-norms.

2. Let $f : [0, 1] \rightarrow [0, \infty]$. Define

$$T_f(x, y) = f^{-1}(f(x) + f(y)), \quad x, y \in [0, 1]. \quad (1)$$

- (a) If $f(x) = (-\ln x)^\lambda$, where $\lambda \in (0, \infty)$, then determine T_f .
- (b) If $f(x) = \left(\frac{1-x}{x}\right)^\lambda$, where $\lambda \in (0, \infty)$, then can you still determine T_f .
- (c) Based on the above examples, can you determine the conditions on an f that would make T_f a t-norm?

3. Consider the function

$$T_Y^w(x, y) = 1 - \min\left(1, \left[(1-a)^w + (1-b)^w\right]^{\frac{1}{w}}\right), \quad x, y \in [0, 1].$$

- (a) Show that for each $w \in (0, \infty]$, T_Y^w is a t-norm.
 - (b) Show that $\lim_{w \rightarrow 0} T_Y^w = T_{\mathbf{D}}$, the drastic t-norm.
 - (c) Show that $\lim_{w \rightarrow \infty} T_Y^w = T_{\mathbf{M}} = \min$.
4. **Lipschitz Continuity of t-norm:** A t-norm $T(x, y)$ is said to be Lipschitz continuous if $\forall x_1, x_2, y_1, y_2 \in [0, 1] \exists k \in (0, \infty)$ such that

$$|T(x_1, y_1) - T(x_2, y_2)| \leq k(|x_1 - x_2| + |y_1 - y_2|).$$

Check whether the t-norm defined as $T(x, y) = \sqrt{\max\{x^2 + y^2 - 1, 0\}}$ is Lipschitz continuous or not.

5. **Left Continuity of t-norm:** A t-norm $T(x, y)$ is said to be left continuous if for any increasing sequence $\{x_n\} \in [0, 1]^{\mathbb{N}}$ and a fixed y_0 ,

$$\sup\{T(x_n, y_0) \mid n \in \mathbb{N}\} = T(\sup\{x_n \mid n \in \mathbb{N}\}, y_0)$$

In other words, for every $x_0 \in [0, 1]$, if

$$\lim_{x \rightarrow x_0^-} T(x, y) = T(x_0, y),$$

then T is left-continuous.

Determine which of the above t-norms is/are left continuous.